

Bitcoin Market Sentiment vs Trader Performance Analysis

Submitted as Part of Data Science Hiring Assessment

Executive Summary

This project analyzes the relationship between Bitcoin market sentiment and trader performance using two datasets: the Bitcoin Fear & Greed Index and Hyperliquid Historical Trader Data. The objective was to determine how market sentiment influences trading profitability, trader behavior, win rates, and trading volume.

The analysis revealed that market sentiment has a significant impact on trading outcomes. Extreme Greed generated the highest average profit per trade and the highest win rate, while Fear generated the largest total profit and trading volume. Extreme Fear was associated with the weakest trader performance.

These findings indicate that sentiment analysis can be used as an additional signal for improving trading strategies and risk management decisions.

Problem Statement

Financial markets are highly influenced by investor psychology and sentiment. The objective of this project is to investigate whether Bitcoin market sentiment, represented by the Fear & Greed Index, affects trader performance on the Hyperliquid platform.

The key questions addressed are:

1. Does trader profitability vary across market sentiment conditions?
 2. Which sentiment generates the highest win rate?
 3. How does trader behavior change during Fear and Greed periods?
 4. Can sentiment data be used to improve trading strategies?
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Dataset Description

Dataset 1: Bitcoin Fear & Greed Index

Columns:

- Timestamp
- Value
- Classification
- Date

Sentiment Categories:

- Extreme Fear
 - Fear
 - Neutral
 - Greed
 - Extreme Greed
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Dataset 2: Hyperliquid Historical Trader Data

Columns:

- Account
- Coin
- Execution Price
- Size Tokens
- Size USD
- Side
- Timestamp
- Start Position
- Direction
- Closed PnL
- Transaction Hash
- Order ID

This dataset contains detailed historical trading records including trader actions, trade sizes, and profit/loss values.

Methodology

Step 1: Data Loading

Both datasets were imported using Python and Pandas.

Step 2: Data Cleaning

- Column names standardized
- Missing values identified
- Data types corrected

Step 3: Date Standardization

Trading timestamps and sentiment dates were converted into a common date format.

Step 4: Data Integration

Both datasets were merged using the date field to associate each trade with the corresponding market sentiment.

Step 5: Exploratory Data Analysis

Analysis was performed on:

- Profitability
- Win Rate
- Trading Volume
- Buy/Sell Behavior
- Trader Segmentation
- Volatility

Exploratory Data Analysis

Sentiment Distribution

Market sentiment records were categorized into:

- Extreme Fear
- Fear
- Neutral
- Greed

- Extreme Greed

The merged dataset allowed sentiment-based performance evaluation.

Average Profitability Analysis

Average Closed PnL by Sentiment:

Sentiment	Average PnL
Extreme Greed	205.82
Fear	128.29
Greed	53.99
Neutral	27.09
Extreme Fear	1.89

Extreme Greed generated the highest average profit per trade.

Total Profitability Analysis

Sentiment	Total PnL
Fear	1.78 Million
Extreme Greed	1.16 Million
Greed	0.61 Million
Neutral	0.07 Million
Extreme Fear	0.004 Million

Fear generated the highest cumulative profitability.

Win Rate Analysis

Sentiment	Win Rate (%)
Extreme Greed	55.33
Neutral	49.49
Greed	43.57

Sentiment	Win Rate (%)
Fear	38.18
Extreme Fear	29.28

Extreme Greed achieved the highest win rate.

Trading Volume Analysis

Sentiment	Trading Volume (USD)
Fear	79.67 Million
Greed	57.05 Million
Extreme Greed	18.22 Million
Neutral	11.94 Million
Extreme Fear	9.58 Million

Fear periods showed the highest trading activity.

Buy vs Sell Behavior

Observations:

- Extreme Greed contained significantly more SELL trades than BUY trades.
- Traders appeared to lock in profits during Greed periods.
- Fear periods showed balanced trading activity.

Volatility Analysis

Profit volatility was measured using standard deviation of Closed PnL.

Sentiment	Standard Deviation
Extreme Greed	1861.56
Greed	1399.47
Fear	1342.35
Neutral	142.95

Sentiment	Standard Deviation
Extreme Fear	76.73

Extreme Greed showed the highest volatility, indicating higher risk and reward.

Key Findings

Finding 1

Extreme Greed produced the highest average profit per trade.

Finding 2

Fear generated the highest cumulative profitability across all trades.

Finding 3

Extreme Greed achieved the highest trader win rate.

Finding 4

Extreme Fear was the least favorable trading environment.

Finding 5

Fear generated the highest trading volume.

Finding 6

SELL orders dominated during Greed and Extreme Greed periods.

Finding 7

Extreme Greed was associated with the highest market volatility.

Recommendations

Recommendation 1

Use market sentiment as an additional trading signal rather than relying solely on price action.

Recommendation 2

Apply stricter risk management during Extreme Fear periods.

Recommendation 3

Monitor Extreme Greed conditions carefully because they offer high profitability but also increased volatility.

Recommendation 4

Use sentiment-aware position sizing and leverage management.

Recommendation 5

Combine sentiment indicators with technical analysis and market structure signals.

Conclusion

The study demonstrates a clear relationship between Bitcoin market sentiment and trader performance. Extreme Greed produced the highest average profitability and win rates, while Fear generated the highest overall profits and trading volume. Extreme Fear was associated with poor trading outcomes.

The results suggest that sentiment-based analysis can improve trading strategies, optimize risk management, and enhance decision-making in cryptocurrency markets.

Tools and Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
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Data Science Hiring Assessment Submission